

The finite volume method solution of the radiative transfer equation for photon transport in non-thermal gas discharges: application to the calculation of photoionization in streamer discharges

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Abstract

This paper presents the development of a direct accurate numerical method to solve the monochromatic radiative transfer equation (RTE) based on a finite volume method (FVM) and its application to the simulation of streamer propagation. The validity of the developed model is demonstrated by performing direct comparisons with results obtained using the classic integral model. Comparisons with approximate solutions of the RTE (Eddington and SP₃ models) are also carried out. Specific validation comparisons are presented for an artificial source of radiation with a Gaussian shape. The reported results demonstrate that whatever the value of the absorption coefficient, the results obtained using the direct FVM are in excellent agreement with the reference integral model with a significantly reduced computation time. When the absorption coefficient is high enough, the Eddington and SP₃ methods are as accurate and become faster than the FVM. However, when the absorption coefficient decreases, approximate methods become less accurate and more computationally expensive than the FVM. Then the direct finite volume and the SP₃ models have been applied to the calculation of photoionization in a double-headed streamer at ground pressure. For high values of the absorption coefficient, positive and negative streamers calculated using the SP₃ model and the FVM for the photoionization source term are in excellent agreement. As the value of the absorption coefficient decreases, discrepancies between the results obtained with the finite volume and the SP₃ models increase, and these differences increase as the streamers advance. For low values of the absorption coefficient, the use of the SP₃ model overestimates the electron density and underestimates the photoionization source term in both streamers in comparison with the FVM. As a consequence, for low values of the absorption coefficient, positive and negative streamers calculated using the SP₃ model for the photoionization source term propagate more slowly than those calculated using the FVM.

(Some figures in this article are in colour only in the electronic version)

1. Introduction

A streamer discharge is a thin weakly ionized plasma filament which propagates very quickly between high voltage

electrodes. For streamer discharges in air at atmospheric pressure (Bastien and Marode 1979, Raizer 1991, Kulikovskiy 2000), the characteristic radius of a streamer discharge is on the order of 20–100 μm , the electron density in the

filament is on the order of 10^{13} – 10^{14} cm $^{-3}$ and the propagation velocity of the streamer is in the range 10^7 – 10^8 cm s $^{-1}$. This filament is strongly non-uniform and consists of a quasi-neutral conducting channel with a weak electric field (on the order of 5–10 kV cm $^{-1}$) and of a very luminous streamer head where the electric field is high (of about 150 kV cm $^{-1}$). The charge in the head can be either positive or negative and the streamer is called a positive or a negative streamer, respectively. The positive streamer moves in the opposite direction to the natural motion of electrons under the action of the electric field. This propagation is the consequence of the existence of an ionizing wave induced by the high value of the electric field in the streamer head. However, for the ionization wave to propagate, a sufficient number of electrons have to be located close to the streamer head (Loeb and Meek 1940). This is the case, for example, in repetitive discharges with a sufficiently high repetition rate. In these conditions, the gas can be efficiently pre-ionized by the previous discharge (Pancheshnyi 2005, Naidis 2006) and then the positive streamer propagates easily in the gas. When the pre-ionization of the gas is too low, the photoionization process is believed to play a critical role in the spatial advancement of positive streamers (e.g. Kunhardt and Tzeng 1988, Babaeva and Naidis 1997, Rocco *et al* 2002, Kulikovskiy 2000, Pancheshnyi 2001, Luque *et al* 2007, Bourdon *et al* 2007). Conversely, the negative streamer propagates in the same direction as the natural motion of electrons. Then photoionization or gas pre-ionization has a smaller influence on the velocity and the radial expansion of the negative streamer in comparison with the positive one.

It is important to note that photoionization is not a unique process in which photons play a significant role for streamer discharges. Indeed, photons emitted by the streamer head may also lead to secondary electron production close to surfaces due to photoemission (e.g. Wu and Kunhardt 1988). Furthermore, in many other types of discharges (such as plasma display panels and lamp discharges), it is necessary to take into account radiative transfer. In general, the radiative transfer equation (RTE) has to be solved in strong connection with the equations of the discharge. For example, in noble gases, when the trapping of resonance radiation has to be considered, space and time variations of resonant excited atom densities have to be known. Then the RTE has to be solved as the photon distribution characterizes the production and loss of excited atoms.

In these latter cases and many others, the main difficulty is to solve the RTE. Very often, a direct integration method is used. With this approach, the determination of the photon distribution function at a given point requires a quadrature over the whole volume of the discharge. Then this calculation in every cell and for every time step in two-dimensional (2D) simulations is computationally expensive, and the problem is obviously worse in three-dimensional (3D) simulations. For the calculation of the photoionization source term in streamer discharges, different methods have been proposed in the literature (see Bourdon *et al* (2007) and references therein) to optimize the calculation of the integral model. These methods reduce the computation time to a certain degree, but they have to be adapted and tested carefully for each new studied

configuration and furthermore they are not straightforward to extend to 3D simulations.

Recently, two different approaches have been proposed to avoid the calculation of the global quadrature over the simulation domain. The first approach (Ségur *et al* 2006) is based on first order (also called the Eddington approximation) and third order (also called the SP $_3$ method) approximations of the RTE. In Bourdon *et al* (2007), we have extended these models to take into account the spectral dependence of photoionization in air and developed the three-group Eddington and SP $_3$ methods. The second approach has been proposed by Luque *et al* (2007) and improved in Bourdon *et al* (2007). This approach is based on an approximation of the absorption function of the gas in order to transform the integral expression of the photoionization term into a set of Helmholtz equations. In Bourdon *et al* (2007) and Liu *et al* (2007), we have successfully applied these differential methods to the propagation in air at atmospheric pressure of a double-headed streamer in a strong electric field and of a positive streamer in a weak field, respectively. In both cases, results have been compared with those obtained using the reference integral photoionization model in air proposed by Zheleznyak *et al* (1982). An excellent agreement was obtained for the three-group SP $_3$ method.

It is important to note that Eddington and SP $_3$ methods are only approximations of the initial RTE. Then their validity range is limited to high values of the absorption coefficient (see Ségur *et al* (2006)). In order to carefully check their range of validity together with their computational efficiency, in this paper we propose to develop a direct accurate numerical solution of the RTE.

In section 2, we present the equations needed for the simulation of streamer discharges. Then we discuss the different integral and differential methods used to solve the monochromatic RTE. In particular, we present the direct finite volume solution of the RTE and we discuss its application to streamer discharge simulations. Finally, we compare the results obtained with the different methods for an artificial Gaussian source of radiation and for a realistic problem involving the development of a double-headed streamer.

2. Model formulation

2.1. Equations for charged and neutral particles

The classical model to simulate the propagation of a streamer discharge is based on the following drift–diffusion equations for electrons and ions coupled with the Poisson's equation (e.g. Dhali and Williams 1987, Kulikovskiy 1997):

$$\frac{\partial n_e}{\partial t} + \vec{\nabla} \cdot n_e \vec{v}_e - \vec{\nabla} \cdot (\vec{D}_e \vec{\nabla} \cdot n_e) = S_{ph} + S_e^+ - S_e^-, \quad (1)$$

$$\frac{\partial n_p}{\partial t} + \vec{\nabla} \cdot n_p \vec{v}_p - \vec{\nabla} \cdot (\vec{D}_p \vec{\nabla} \cdot n_p) = S_{ph} + S_p^+ - S_p^-, \quad (2)$$

$$\frac{\partial n_n}{\partial t} + \vec{\nabla} \cdot n_n \vec{v}_n - \vec{\nabla} \cdot (\vec{D}_n \vec{\nabla} \cdot n_n) = S_n^+ - S_n^-, \quad (3)$$

$$\nabla^2 V = -\frac{q_e}{\epsilon_0}(n_p - n_n - n_e), \quad (4)$$

where subscripts e , p and n refer to electrons, positive and negative ions, respectively. n_i is the number density of species i , V is the potential, $\vec{v}_i = \mu_i \vec{E}$ ($\vec{E}$ being the electric field) is the drift velocity of species i and $\vec{D}_i$ and μ_i are the diffusion tensor and the mobility of species i , respectively; q_e and ϵ_0 are the absolute value of the electron charge and the permittivity of free space. The S^+ and S^- terms are the rates of production and loss of charged particles. The S_{ph} term is the rate of electron-ion pair production due to photoionization in the gas.

As the production and loss of charged particles are due to many physico-chemical processes, it is also necessary to write the balance equations for neutral species in fundamental and excited states. These equations can be written under the same general form. For example, for an atom or a molecule on an excited (rotational, vibrational or electronic) level u , we have (Yoshida and Tagashira 1976)

$$\frac{\partial n_u}{\partial t} - D_u \nabla^2 n_u = \nu_u n_e - \sum_{d < u} \frac{n_u}{\tau_{ud}} + \sum_{d > u} \frac{n_d}{\tau_{du}} - S_u^- + S_u^+ + S_{ph}^{\text{exc}}, \quad (5)$$

where D_u is the diffusion coefficient of the excited atom or molecule on level u in the gas and ν_u is the electron impact excitation frequency of the atom or molecule on level u . The second term on the right-hand side of equation (5) represents the radiative deexcitation of level u on different lower levels $d < u$ and $1/\tau_{ud}$ is the Einstein coefficient for the spontaneous transition $u \rightarrow d$. The third term is a source term due to radiative deexcitation of upper levels $d > u$ on level u , S_u^- is the loss term due to the quenching of level u and S_u^+ is the source term due to quenching of upper levels $d > u$. Finally, the production of an excited atom or molecule on level u by photons is described by the photoexcitation source term S_{ph}^{exc} .

2.2. Radiative source terms and the radiative transfer equation

The photoionization source term $S_{ph}(\vec{r}, t)$ at position $\vec{r}$ and at time t in equations (1) and (2) is related directly to the photon distribution function $\Psi_v(\vec{r}, \vec{\Omega}, t)$ of frequency ν at position $\vec{r}$ in the direction $\vec{\Omega}$ and at time t , according to the following relation:

$$\begin{aligned} S_{ph}(\vec{r}, t) &= c \int_0^\infty d\nu \xi_v^{\text{ph}} \mu_{\text{abs},v} \int_{\Omega} d\Omega \Psi_v(\vec{r}, \vec{\Omega}, t) \\ &= c \int_0^\infty d\nu \xi_v^{\text{ph}} \mu_{\text{abs},v} \Psi_{0,v}(\vec{r}, t), \end{aligned} \quad (6)$$

where c is the speed of light, $\mu_{\text{abs},v}$ is the absorption coefficient which depends on the frequency ν , $0 \leq \xi_v^{\text{ph}} \leq 1$ is the photoionization efficiency which is equal to the ratio of the number of photoelectrons appearing to the total number of absorbed photons of frequency ν (Zheleznyak *et al* 1982) and $\Psi_{0,v}(\vec{r}, t)$ is the isotropic part of the photon distribution function for a given frequency ν defined by

$$\Psi_{0,v}(\vec{r}, t) = \int_{\Omega} d\Omega \Psi_v(\vec{r}, \vec{\Omega}, t). \quad (7)$$

Obviously, in equation (6), to calculate the total photoionization source term, a quadrature over the frequency range is required. Similarly to equation (6), the photoexcitation source term on level u , $S_{ph}^{\text{exc}}(\vec{r}, t)$, in equation (5) can be written as (Molisch and Oehrig 1998)

$$\begin{aligned} S_{ph}^{\text{exc}}(\vec{r}, t) &= c \sum_{d < u} B_{du} \int_0^\infty d\nu \mu_{\text{abs},v} \int_{\Omega} d\Omega \Psi_v(\vec{r}, \vec{\Omega}, t) \\ &= c \sum_{d < u} B_{du} \int_0^\infty d\nu \mu_{\text{abs},v} \Psi_{0,v}(\vec{r}, t), \end{aligned} \quad (8)$$

where B_{du} is the Einstein absorption coefficient. The photon distribution function $\Psi_v(\vec{r}, \vec{\Omega}, t)$ is given by the RTE:

$$\begin{aligned} \frac{\partial \Psi_v(\vec{r}, \vec{\Omega}, t)}{\partial t} + c \vec{\Omega} \cdot \vec{\nabla} \Psi_v(\vec{r}, \vec{\Omega}, t) \\ = \sum_{u,d} \frac{n_u(\vec{r}, t) \phi_{ud}(\nu)}{4\pi \tau_{ud}} - c \mu_{\text{abs},v} \Psi_v(\vec{r}, \vec{\Omega}, t), \end{aligned} \quad (9)$$

where $n_u(\vec{r}, t)$ is the density of the radiative excited species u at position $\vec{r}$ and time t (given by equation (5)) and $\int_0^\infty \phi_{ud}(\nu) d\nu = 1$ is the normalized emission line profile for the spontaneous transition $u \rightarrow d < u$. The summation over u and d in the first term on the right-hand side of equation (9) means that all transitions contributing to frequency ν are taken into account. In equation (9), the scattering and the change in frequency of photons during collisions with molecules are neglected (Goody 1996). Then a photon, once emitted, moves along a straight line before being absorbed.

Within the time scale of streamer propagation (a few tens of nanoseconds in centimetre gaps), photon propagation is nearly instantaneous and then the transient term in equation (9) is neglected. Furthermore, if we assume, as proposed by Zheleznyak *et al* (1982), that the radiative excited species are in steady state (i.e. the transient term in equation (5) is neglected), equation (9) becomes

$$\vec{\Omega} \cdot \vec{\nabla} \Psi_v(\vec{r}, \vec{\Omega}) = \sum_{u,d} \frac{n_u(\vec{r}) \phi_{ud}(\nu)}{4\pi c \tau_{ud}} - \mu_{\text{abs},v} \Psi_v(\vec{r}, \vec{\Omega}), \quad (10)$$

In equation (10) and the following, to simplify notations, the time variable t is omitted. In the following, our objective is mainly to check the efficiency of various numerical methods for solving equation (10). Then, to simplify, we consider a monochromatic approach and we write $\Psi_v(\vec{r}, \vec{\Omega}) = \Psi(\vec{r}, \vec{\Omega}) \delta(\nu)$ where $\delta(\nu)$ is the Dirac function. After integration of equation (10) over the frequency ν , we obtain

$$\vec{\Omega} \cdot \vec{\nabla} \Psi(\vec{r}, \vec{\Omega}) = \frac{Q(\vec{r})}{4\pi} - \mu_{\text{abs}} \Psi(\vec{r}, \vec{\Omega}), \quad (11)$$

where μ_{abs} is the monochromatic absorption coefficient and

$$Q(\vec{r}) = \frac{n_u(\vec{r})}{c \tau_u}, \quad (12)$$

where to simplify, only one excited state u with its radiative relaxation time τ_u is taken into account. In the model proposed by Zheleznyak *et al* (1982) for photoionization in air, equation (5) is simplified to derive the ratio $n_u(\vec{r})/\tau_u$ as

$$\frac{n_u(\vec{r})}{\tau_u} = \frac{p_q}{p + p_q} \frac{\nu_u}{\nu_i} v_i(\vec{r}) n_e(\vec{r}), \quad (13)$$

where the ratio $p_q/(p + p_q)$ is a quenching factor and ν_i is the electron impact ionization frequency. Finally, for a monochromatic approach, the photoionization source term given by equation (6) becomes

$$S_{\text{ph}}(\vec{r}) = c\xi^{\text{ph}}\mu_{\text{abs}}\Psi_0(\vec{r}), \quad (14)$$

where ξ^{ph} is the monochromatic photoionization efficiency.

2.3. Numerical solution of the radiative transfer equation: integral methods

As equation (11) is a first order partial differential equation, its integration over the whole space (Pomraning 1973, Lewis and Miller 1984) and over the solid angle $d\Omega$ is straightforward. It gives the isotropic part $\Psi_0(\vec{r})$ of the distribution function at a given point $\vec{r}$:

$$\Psi_0(\vec{r}) = \int_{\text{Vol}} d^3r' \frac{Q(\vec{r}')}{4\pi R^2} \exp(-\mu_{\text{abs}}R), \quad (15)$$

where $R = |\vec{r} - \vec{r}'|$. Equation (15) simply states that the photon distribution at a given point $\vec{r}$ corresponds to the total fraction of photons emitted isotropically at every point $\vec{r}'$ in the gap and which are not absorbed after a distance of $|\vec{r} - \vec{r}'|$. To calculate the integral in equation (15), standard quadrature methods can be used. However, the calculation time is very long as (i) this is a quadrature on the whole volume which has to be done for each point inside the computational domain and (ii) for the calculation of the photoionization source term in streamer discharges, this quadrature has to be done at every time step. Different methods have been proposed in the literature to decrease the computational cost of the integral model (see Bourdon *et al* (2007) and references therein). For example, Kulikovskiy (2000) and Hallac *et al* (2003) propose to calculate the photoionization source term on a coarse grid and interpolation is used to obtain the needed values on the main grid. In Pancheshnyi *et al* (2001), calculations are limited to a small area around the streamer head.

2.4. Numerical solution of the radiative transfer equation: approximate differential methods

Usually, in the literature, as the direct numerical solution of equation (11) is not straightforward, approximate differential methods are used. These methods are based on the assumption that the distribution function $\Psi(\vec{r}, \vec{\Omega})$ varies only weakly with the unitary vector $\vec{\Omega}$. Using this assumption, equation (11) is transformed into a partial differential equation of second order which gives directly the isotropic part $\Psi_0(\vec{r})$. This is the so-called Eddington approximation (Pomraning 1973). Unfortunately, this approximation is valid only if the absorption coefficient μ_{abs} is sufficiently high (i.e. in a strongly absorbing medium). To obtain higher order approximations, Larsen *et al* (2002) proposed to introduce a small parameter in equation (11) in order to expand this equation in a series of powers of this small parameter. Following this idea, we

propose to divide equation (11) by μ_{abs} and to define the vector $\vec{\mathfrak{R}} = \vec{\nabla}/\mu_{\text{abs}}$. Then, equation (11) becomes

$$(1 + \vec{\Omega} \cdot \vec{\mathfrak{R}})\Psi(\vec{r}, \vec{\Omega}) = \frac{Q(\vec{r})}{4\pi\mu_{\text{abs}}}. \quad (16)$$

Equation (16) shows that when the parameter $0 < \vec{\Omega} \cdot \vec{\mathfrak{R}} \ll 1$, it is possible to apply Neumann's series. After an integration over the solid angle $d\Omega$, finally, we obtain the generalized Eddington approximation (Larsen *et al* 2002, Ségur *et al* 2006):

$$\left(1 - \frac{1}{3\mu_{\text{abs}}^2}\nabla^2 - \frac{4}{45\mu_{\text{abs}}^4}\nabla^4 - \frac{44}{945\mu_{\text{abs}}^6}\nabla^6 + \dots\right)\Psi_0(\vec{r}) = \frac{Q(\vec{r})}{\mu_{\text{abs}}}. \quad (17)$$

If we consider only the second order terms in this expansion, we obtain exactly the Eddington approximation, i.e.

$$[\nabla^2 - 3\mu_{\text{abs}}^2]\Psi_0(\vec{r}) = -3\mu_{\text{abs}}Q(\vec{r}). \quad (18)$$

Boundary conditions for the Eddington approximation have been discussed in Pomraning (1973). In Larsen *et al* (2002), a significant improvement of the accuracy of the solution was obtained by keeping the terms on the left-hand side of equation (17) up to the third order Laplacian. In this case, the equation is a partial differential equation of sixth order which can be written as a set of two equations of second order verified by the functions ϕ_1 and ϕ_2 and given by

$$\nabla^2\phi_1(\vec{r}) - \frac{\mu_{\text{abs}}^2}{\kappa_1^2}\phi_1(\vec{r}) = -\frac{\mu_{\text{abs}}}{\kappa_1^2}Q(\vec{r}), \quad (19)$$

$$\nabla^2\phi_2(\vec{r}) - \frac{\mu_{\text{abs}}^2}{\kappa_2^2}\phi_2(\vec{r}) = -\frac{\mu_{\text{abs}}}{\kappa_2^2}Q(\vec{r}), \quad (20)$$

where κ_1^2 and κ_2^2 are constants (see Larsen *et al* (2002)). In the following, the third order approximation is called SP₃, as in Larsen *et al* (2002). The SP₃ approximation of Ψ_0 is denoted as $\Psi_{\text{SP}_3,0}$ and is given by

$$\Psi_{\text{SP}_3,0}(\vec{r}) = \frac{\gamma_2\phi_1(\vec{r}) - \gamma_1\phi_2(\vec{r})}{\gamma_2 - \gamma_1}, \quad (21)$$

where γ_n are constants (see Larsen *et al* (2002)). The two elliptic equations (19) and (20) are loosely coupled by their boundary conditions (Larsen *et al* (2002)).

2.5. Numerical solution of the radiative transfer equation: direct methods

In the following, we limit our investigation to a two-dimensional axisymmetric cylindrical discharge. In this case, the photon distribution depends only on two space variables (z and r) and two angles (θ and φ) defined in figure 1. z and r are the longitudinal and radial positions and θ and φ are the polar and azimuthal angles and define the vector $\vec{\Omega}$ in the unit sphere. Variables μ ($= \sin\theta \cos\varphi$), η ($= \sin\theta \sin\varphi$) and ξ ($= \cos\theta$) are direction cosines. Then, equation (11) can be written as (Lewis and Miller 1984, Chui and Raithby 1992):

$$\left[\mu\frac{\partial}{\partial r} + \xi\frac{\partial}{\partial z} - \frac{\eta}{r}\frac{\partial}{\partial\varphi} + \mu_{\text{abs}}\right]\Psi(z, r, \theta, \varphi) = \frac{Q(z, r)}{4\pi}. \quad (22)$$

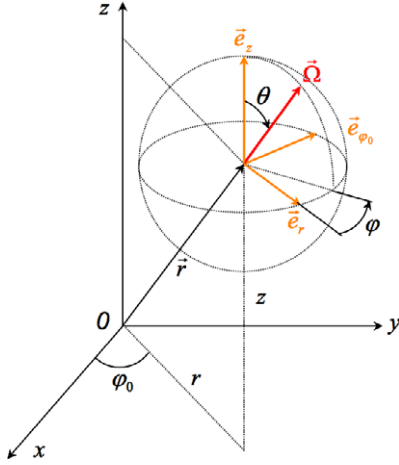


Figure 1. Definition of the different variables used for equation (22).

It is interesting to note that for the radiative transfer in a streamer discharge at atmospheric pressure, the scattering and the change in frequency of photons during collisions with atoms and molecules have been neglected in equation (22) (Goody 1996). For applications in which the scattering effect is important, an additional source term has to be taken into account in equation (22). In this case, this equation becomes an integro-differential equation and the numerical method used to solve it becomes iterative. Different numerical methods exist in the literature to solve equation (22). One widely used is the S_N method which was originally developed by Carlson and Lathrop (1968) to solve the equation of neutron transport and later extended to radiative transfer problems. The principle of the S_N method is to use the discrete ordinate method on the angular variable of equation (22) followed by an application of the control volume method on space and energy (frequency in the case of photons) variables. Details on the various techniques available can be found in Lee (1962), Lathrop (1966) and Modest (2003). The S_N method gives accurate numerical results in the case of planar and spherical geometries when it can be assumed that the distribution function depends only on one space variable. In the case of more complex geometries (cylindrical geometry, for example) or when it is necessary to take into account a higher number of space variables, the numerical results are not very accurate. In particular, a phenomenon called the ‘ray effect’ appears, resulting in oscillations in the angular dependence of the distribution function. The existence of this effect is directly related to the order of the angular discretization used in the S_N method. Then this effect can be reduced by using a higher order of discretization, but this may significantly increase the computation time. However, it is important to note that the main shortcoming of the S_N method is that it does not guarantee the conservation of energy radiation. This is again the result of the use of the discrete ordinate method for the discretization of the angular distribution function.

To eliminate the different problems mentioned above, currently, for many applications, the finite volume method (FVM) is used not only on space variables but also on angular variables (Chui and Raithby 1992). To solve equation (22)

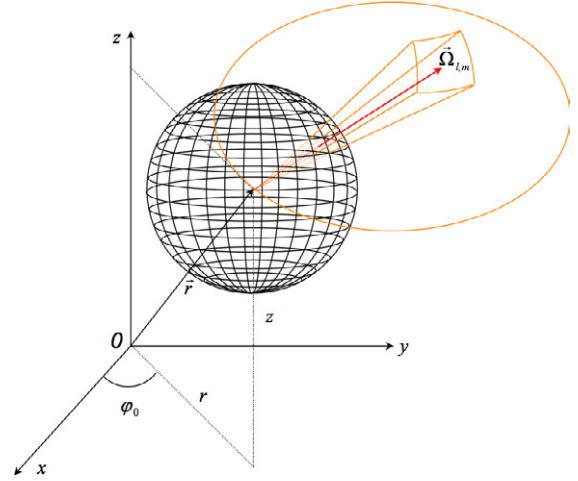


Figure 2. Distribution chosen for the elementary solid angles on the unit sphere and definition of the direction $\vec{\Omega}_{l,m}$.

using the FVM, first, we write this equation under the so-called conservative form. We have (Lewis and Miller 1984)

$$\left[\frac{\mu}{r} \frac{\partial}{\partial r} r + \xi \frac{\partial}{\partial z} - \frac{1}{r} \frac{\partial}{\partial \varphi} \eta + \mu_{\text{abs}} \right] \Psi(z, r, \theta, \varphi) = \frac{Q(z, r)}{4\pi}. \quad (23)$$

The interest in this form is that the basic physical conservative properties of the initial RTE clearly appear. The FVM is based on the integration of equation (23) over an elementary volume element $r dr dz d\varphi_0 \sin \theta d\theta d\varphi$. To do that, we define a set of grids both in angles and space variables. Figure 2 shows the uniform distribution chosen for the solid angles on the unit sphere. The elementary solid angle $d\Omega_{l,m}$ (with $l = 1, 2, \dots, N_\theta$; $m = 1, 2, \dots, N_\varphi$) is centred on the direction $\vec{\Omega}_{l,m}$ which is defined by the polar and azimuthal angles θ_l and φ_m , respectively. The boundaries of this solid angle are between $\theta_{l-1/2}$ and $\theta_{l+1/2}$ and between $\varphi_{m-1/2}$ and $\varphi_{m+1/2}$. We have also to divide the z and r axes into a certain number of volume elements centred on the points z_i and r_j (with $i = 1, 2, \dots, N_z$; $j = 1, 2, \dots, N_r$). The volume $V_{i,j}$ of every element is given by

$$V_{i,j} = \pi(r_{j+1/2}^2 - r_{j-1/2}^2) \Delta z_i = A_j \Delta z_i, \quad (24)$$

where $\Delta z_i = z_{i+1/2} - z_{i-1/2}$ and A_j is an annular surface centred on the circle of radius r_j . As the geometry is axisymmetric, the result of the integration of equation (23) over the elementary volume element $r dr dz d\varphi_0 \sin \theta d\theta d\varphi$ is independent of the angle φ_0 and then is given by

$$\int_{z_{i-1/2}}^{z_{i+1/2}} dz \int_{r_{j-1/2}}^{r_{j+1/2}} r dr \int_{\theta_{l-1/2}}^{\theta_{l+1/2}} \sin \theta d\theta \int_{\varphi_{m-1/2}}^{\varphi_{m+1/2}} d\varphi \times \left[\frac{\mu}{r} \frac{\partial}{\partial r} (r\Psi) + \xi \frac{\partial \Psi}{\partial z} - \frac{1}{r} \frac{\partial}{\partial \varphi} (\eta\Psi) + \mu_{\text{abs}} \Psi - \frac{Q}{4\pi} \right] = 0, \quad (25)$$

which becomes

$$\begin{aligned}
& 2\pi \bar{\mu}_{l,m} \Delta z_i [r_{j+1/2} \bar{\Psi}_{i,j+1/2,l,m} - r_{j-1/2} \bar{\Psi}_{i,j-1/2,l,m}] \\
& + \bar{\xi}_{l,m} A_j [\bar{\Psi}_{i+1/2,j,l,m} - \bar{\Psi}_{i-1/2,j,l,m}] \\
& + 2\pi \Delta r_j \Delta z_i [\alpha_{m+1/2}^l \bar{\Psi}_{i,j,l,m+1/2} - \alpha_{m-1/2}^l \bar{\Psi}_{i,j,l,m-1/2}] \\
& + \mu_{\text{abs}} V_{i,j} \bar{\Psi}_{i,j,l,m} = V_{i,j} \frac{\bar{Q}_{i,j}}{4\pi}, \quad (26)
\end{aligned}$$

where $\bar{\mu}_{l,m}$ and $\bar{\xi}_{l,m}$ are mean values of μ and ξ on the elementary solid angle $d\Omega_{l,m}$ and are defined by

$$\bar{\mu}_{l,m} = \frac{D_{l,m}^1}{D_{l,m}} \quad (27)$$

and

$$\bar{\xi}_{l,m} = \frac{D_{l,m}^2}{D_{l,m}} \quad (28)$$

with

$$\begin{aligned}
D_{l,m}^1 &= \frac{1}{2} [\theta_{l+1/2} - \theta_{l-1/2} - \frac{1}{2} (\sin 2\theta_{l+1/2} - \sin 2\theta_{l-1/2})] \\
&\times (\sin \varphi_{m+1/2} - \sin \varphi_{m-1/2}), \quad (29)
\end{aligned}$$

$$D_{l,m}^2 = -\frac{1}{4} (\cos 2\theta_{l+1/2} - \cos 2\theta_{l-1/2}) (\varphi_{m+1/2} - \varphi_{m-1/2}), \quad (30)$$

$$D_{l,m} = -(\cos \theta_{l+1/2} - \cos \theta_{l-1/2}) (\varphi_{m+1/2} - \varphi_{m-1/2}), \quad (31)$$

The coefficients $\alpha_{m\pm 1/2}^l$ in equation (26) are chosen to satisfy the conservative properties of this equation and are defined by the following recursive relation (Lewis and Miller 1984):

$$\begin{cases} \alpha_{1/2}^l = 0, \\ \alpha_{m+1/2}^l - \alpha_{m-1/2}^l = -\bar{\mu}_{l,m}. \end{cases} \quad (32)$$

Note that the different terms of equation (26) are equivalent to the corresponding terms of equation (23). The main differences are that the continuous derivatives in equation (23) are replaced by differences between surface averaged distribution functions ($\bar{\Psi}_{i,j\pm 1/2,l,m}$, $\bar{\Psi}_{i\pm 1/2,j,l,m}$ and $\bar{\Psi}_{i,j,l,m\pm 1/2}$) in equation (26) and that the distribution function and the source term in equation (23) become mean values ($\bar{\Psi}_{i,j,l,m}$ and $\bar{Q}_{i,j}$, respectively) defined on the whole control volume in equation (26). For example, for the volume averaged distribution function $\bar{\Psi}_{i,j,l,m}$ and the volume averaged source term $\bar{Q}_{i,j}$, we have

$$\begin{aligned}
\bar{\Psi}_{i,j,l,m} &= \frac{2\pi}{D_{l,m} V_{i,j}} \int_{\theta_{l-1/2}}^{\theta_{l+1/2}} \sin \theta d\theta \int_{\varphi_{m-1/2}}^{\varphi_{m+1/2}} d\varphi \\
&\times \int_{r_{j-1/2}}^{r_{j+1/2}} r dr \int_{z_{i-1/2}}^{z_{i+1/2}} dz \Psi(z, r, \theta, \varphi) \quad (33)
\end{aligned}$$

and

$$\bar{Q}_{i,j} = \frac{2\pi}{V_{i,j}} \int_{z_{i-1/2}}^{z_{i+1/2}} dz \int_{r_{j-1/2}}^{r_{j+1/2}} r dr Q(z, r). \quad (34)$$

In the same way, the surface averaged distribution function located at the interface $z_{i+1/2}$ is given by

$$\begin{aligned}
\bar{\Psi}_{i+1/2,j,l,m} &= \frac{2\pi}{D_{l,m} A_j} \int_{\theta_{l-1/2}}^{\theta_{l+1/2}} \sin \theta d\theta \int_{\varphi_{m-1/2}}^{\varphi_{m+1/2}} d\varphi \\
&\times \int_{r_{j-1/2}}^{r_{j+1/2}} r dr \Psi(z_{i+1/2}, r, \theta, \varphi). \quad (35)
\end{aligned}$$

Similar expressions can be easily obtained for the other surface averaged distribution functions $\bar{\Psi}_{i-1/2,j,l,m}$, $\bar{\Psi}_{i,j\pm 1/2,l,m}$ and $\bar{\Psi}_{i,j,l,m\pm 1/2}$. Equation (26) clearly shows that the volume averaged distribution function $\bar{\Psi}_{i,j,l,m}$ depends on the source term $\bar{Q}_{i,j}$ and on the differences between the surface averaged distribution functions at all the input and output faces of the control volume. To calculate $\bar{\Psi}_{i,j,l,m}$, it is then necessary to define a method to determine the different averaged distribution functions. As equation (26) contains seven different values of the distribution function, first their number has to be decreased.

To do that, we have to consider the two situations of photon propagation in a vertical cylindrical domain. The first case is when photons move from the bottom to the top of the cylinder and the second case is the opposite. In the following, we present in detail the procedure for the first case and the guidelines are given for the second case. In the first case, the polar angle θ lies between 0 and $\pi/2$. To solve equation (26), it is necessary to know the value of the distribution function for $z = 0$ whatever the values of θ , r and φ (i.e. $\Psi(0, r, \theta, \varphi)$). This value of Ψ characterizes the distribution of photons coming into the cylinder through its bottom. In our case, it is equal to zero (i.e. no photons are coming from the outside of the domain). The second boundary condition to be known is $\Psi(z, R_d, \theta, \varphi)$, the distribution of photons coming into the cylinder for $r = R_d$ (where R_d is the radius of the cylindrical computational domain) whatever the values of z and $\theta \in [0, \pi/2]$. If r lies between R_d and 0, the azimuthal angle φ lies between π and $\pi/2$. For these latter values, $\Psi(z, R_d, \theta, \varphi)$ is the distribution of photons coming from outside and crossing the external cylinder at $r = R_d$. In our case, this value is equal to zero. Furthermore, we will show in the following that it is possible to determine $\Psi(z, r, \theta, \varphi)$ when $\varphi = \pi$. It is now clear that the functions $\bar{\Psi}_{i,j+1/2,l,m}$, $\bar{\Psi}_{i-1/2,j,l,m}$ and $\bar{\Psi}_{i,j,l,m+1/2}$ are known at the different boundaries. Then, at this step, equation (26) depends only on four values of the distribution function.

As one of these four values is the volume averaged distribution $\bar{\Psi}_{i,j,l,m}$, our objective now is to try to express three surface averaged distribution functions (i.e. $\bar{\Psi}_{i,j-1/2,l,m}$, $\bar{\Psi}_{i+1/2,j,l,m}$ and $\bar{\Psi}_{i,j,l,m-1/2}$) as a function of this volume averaged distribution function and of the other surface averaged distribution functions (i.e. $\bar{\Psi}_{i,j+1/2,l,m}$, $\bar{\Psi}_{i-1/2,j,l,m}$ and $\bar{\Psi}_{i,j,l,m+1/2}$). The simplest way to do that is to assume that the central value $\bar{\Psi}_{i,j,l,m}$ is the weighted average of the surface averaged values at the interfaces as

$$\begin{aligned}
\bar{\Psi}_{i,j,l,m} &= a_1 \times \bar{\Psi}_{i+1/2,j,l,m} + (1 - a_1) \times \bar{\Psi}_{i-1/2,j,l,m} \\
&= (1 - a_2) \times \bar{\Psi}_{i,j+1/2,l,m} + a_2 \times \bar{\Psi}_{i,j-1/2,l,m} \\
&= (1 - a_3) \times \bar{\Psi}_{i,j,l,m+1/2} + a_3 \times \bar{\Psi}_{i,j,l,m-1/2}, \quad (36)
\end{aligned}$$

where a_1 , a_2 and a_3 are constants lying between 0.5 and 1 (Liu *et al* 1996). If we extract $\bar{\Psi}_{i,j-1/2,l,m}$, $\bar{\Psi}_{i+1/2,j,l,m}$ and $\bar{\Psi}_{i,j,l,m-1/2}$ from equation (36) and if we use them in equation (26), we obtain

$$\bar{\Psi}_{i,j,l,m} = \frac{\kappa_{i,j,l,m}}{\lambda_{i,j,l,m}}, \quad (37)$$

where

$$\begin{aligned} \kappa_{i,j,l,m} = & V_{i,j} \frac{\bar{Q}_{i,j}}{4\pi} - 2\pi \Delta z_i \left(r_{j+1/2} + \frac{1-a_2}{a_2} r_{j-1/2} \right) \\ & \times \bar{\mu}_{l,m} \bar{\Psi}_{i,j+1/2,l,m} + \frac{A_j}{a_1} \bar{\xi}_{l,m} \bar{\Psi}_{i-1/2,j,l,m} \\ & - 2\pi \Delta z_i \Delta r_j \left(\alpha_{m+1/2}^l + \frac{1-a_3}{a_3} \alpha_{m-1/2}^l \right) \bar{\Psi}_{i,j,l,m+1/2} \end{aligned} \quad (38)$$

and

$$\begin{aligned} \lambda_{i,j,l,m} = & \mu_{\text{abs}} V_{i,j} - \frac{2\pi \Delta z_i}{a_2} r_{j-1/2} \bar{\mu}_{l,m} + \frac{A_j}{a_1} \bar{\xi}_{l,m} \\ & - \frac{2\pi \Delta z_i \Delta r_j}{a_3} \alpha_{m-1/2}^l. \end{aligned} \quad (39)$$

The recurrence relation (37) allows one to determine the central volume distribution function $\bar{\Psi}_{i,j,l,m}$ for every control volume. Once the central value is calculated, the next surface values $\bar{\Psi}_{i,j+1/2,l,m}$, $\bar{\Psi}_{i-1/2,j,l,m}$ and $\bar{\Psi}_{i,j,l,m+1/2}$ are calculated using the extrapolation relationships given by equation (36). The subsequent central values are then calculated and the process is repeated for every cell. However, some changes have to be done in equations (36)–(39) depending on the values of the angles for which calculations are made. Indeed, relations (36)–(39) are valid for $\theta \in [0, \pi/2]$ and $\varphi \in [\pi/2, \pi]$. In this case, photons move from high to low values of r . For the same values of θ , if $\varphi \in [0, \pi/2]$, photons move from low to high values of r . In this case, in relations (36)–(39), the distribution $\bar{\Psi}_{i,j+1/2,l,m}$ has to be replaced by $\bar{\Psi}_{i,j-1/2,l,m}$ and the coefficients have to be slightly modified to take into account the change in the direction of motion. When $\theta \in [\pi/2, \pi]$, photons move from the top to the bottom of the cylinder, and in equations (36)–(39), the distribution $\bar{\Psi}_{i-1/2,j,l,m}$ has to be replaced by $\bar{\Psi}_{i+1/2,j,l,m}$, and the coefficients have also to be slightly modified to take into account the change in the direction of motion.

Finally, as already mentioned above, we have to determine the boundary condition at $\varphi = \pi$. To do that, we write the RTE equation for this value of φ :

$$\begin{aligned} \mu(\theta) \frac{\partial \Psi}{\partial r}(z, r, \theta) + \xi(\theta) \frac{\partial \Psi}{\partial z}(z, r, \theta) \\ = -\mu_{\text{abs}} \Psi(z, r, \theta) + \frac{Q(z, r)}{4\pi}. \end{aligned} \quad (40)$$

This equation is similar to equation (22) and if we apply the FVM in the same way as for the general case, we obtain a recurrence relationship similar to equation (37), but without the terms corresponding to the partial derivative with respect to φ .

In summary, the general procedure to calculate the photon distribution function with the FVM starts by solving the boundary equation (40). Then, equation (37) is used for

all values of θ and φ . Usually, the parameters a_1 , a_2 and a_3 are taken equal to 0.5 (Liu *et al* 1996). However, in some cases, the extrapolated surface distribution functions calculated using equation (36) may become negative. When this occurs, the parameters a_1 , a_2 and a_3 are fixed to 1. With this choice, Lathrop (1966) has shown that only positive values of surface averaged distribution functions are obtained, but this choice decreases the accuracy of the numerical scheme. It is interesting to note that, due to the cylindrical symmetry, calculations are carried out only inside the range $\varphi \in [0, \pi]$. The values of the distribution function in the range $\varphi \in [\pi, 2\pi]$ are simply obtained by symmetry.

Finally, once the calculations are done for all the values of the central volume distribution functions $\bar{\Psi}_{i,j,l,m}$, the isotropic part $\bar{\Psi}_{0,i,j}$ is given by

$$\bar{\Psi}_{0,i,j} = 2 \sum_{l=1}^{N_\theta} \sum_{m=1}^{N_\varphi} D_{l,m} \bar{\Psi}_{i,j,l,m}. \quad (41)$$

3. Results and discussion

3.1. Test case with a Gaussian source term

In this subsection, a simple time-independent model source of radiation is used to compare the FVM, the Eddington and SP₃ models with the reference integral model for values of the absorption coefficient μ_{abs} ranging from 10 (nearly optically thin case) to 300 cm⁻¹ (nearly optically thick case). We calculate the isotropic part of the photon distribution function $\Psi_0(z, r)$ in a two-dimensional axisymmetric computational domain of length $L_d = 1.4$ cm and radius $R_d = 0.125$ cm for a Gaussian source centred on the symmetry axis defined by

$$Q(z, r) = Q_0 \exp \left[- \left(\frac{z - z_0}{\sigma} \right)^2 - \left(\frac{r}{\sigma} \right)^2 \right], \quad (42)$$

where $Q_0 = 4\pi$ cm⁻⁴, $z_0 = 0.7$ cm is the axial position of the source term and σ is the parameter controlling the effective spatial width of the source. In the following, the value of $\sigma = 0.01$ cm is chosen to be comparable to the size of the streamer head. In this paper, we present the calculations carried out with an exponentially refined grid in both directions (i.e. close to the symmetry axis at $r = 0$ cm and close to the axis $z_0 = 0.7$ cm) with $N_z = 500$ and $N_r = 200$. For the direct method, we have used $N_\theta = N_\varphi = 14$. As part of the preparatory work for the model studies presented in this paper, we have conducted several test runs with different numbers of grid points with refined and uniform meshes. We have checked that the results obtained with the different methods and the mesh used in this paper are the same as those obtained with more refined grids.

Figures 3–5 show the values of the isotropic part of the photon distribution function on the axis of symmetry $\Psi_0(z, r = 0)$ calculated for three values of the absorption coefficient: $\mu_{\text{abs}} = 300$, 100 and 10 cm⁻¹. When the absorption coefficient is high ($\mu_{\text{abs}} = 300$ cm⁻¹), figure 3 shows that the different methods give very similar results. When the value of the absorption coefficient

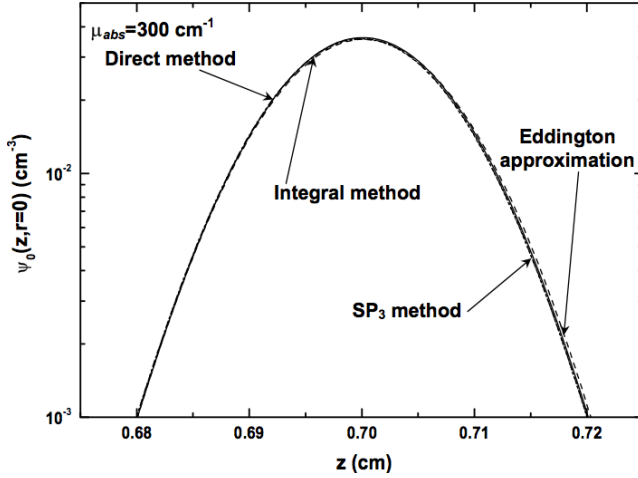


Figure 3. Axial profiles of the isotropic part of the photon distribution function $\Psi_0(z, r = 0)$ for a monochromatic absorption coefficient $\mu_{abs} = 300 \text{ cm}^{-1}$. Dotted-dashed line: SP₃ method; solid line: direct finite volume method; dotted line: Integral method; dashed line: Eddington approximation.

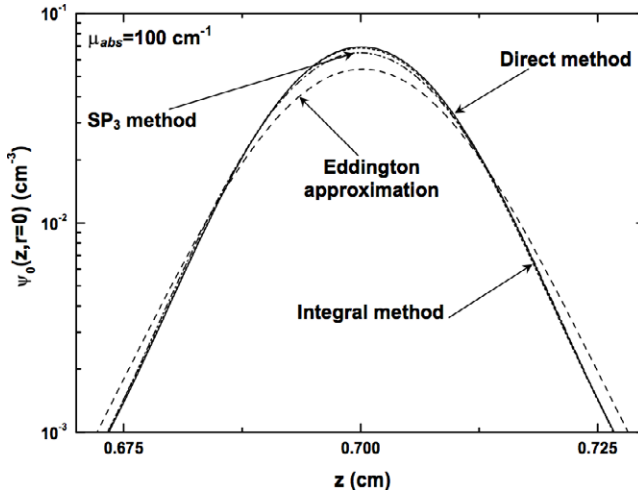


Figure 4. Same caption as figure 3 for a monochromatic absorption coefficient $\mu_{abs} = 100 \text{ cm}^{-1}$.

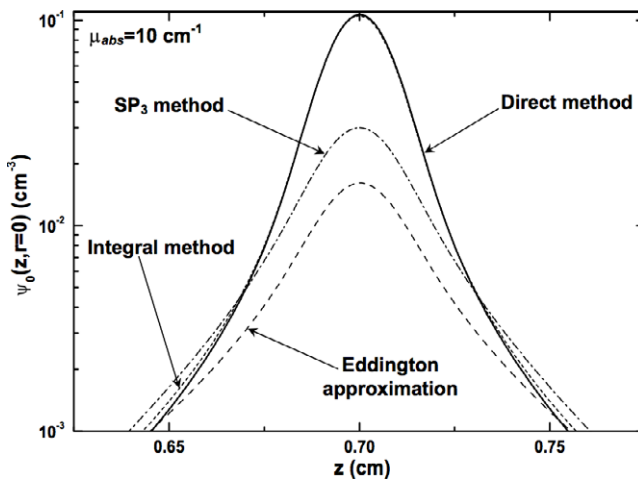


Figure 5. Same caption as figure 3 for a monochromatic absorption coefficient $\mu_{abs} = 10 \text{ cm}^{-1}$.

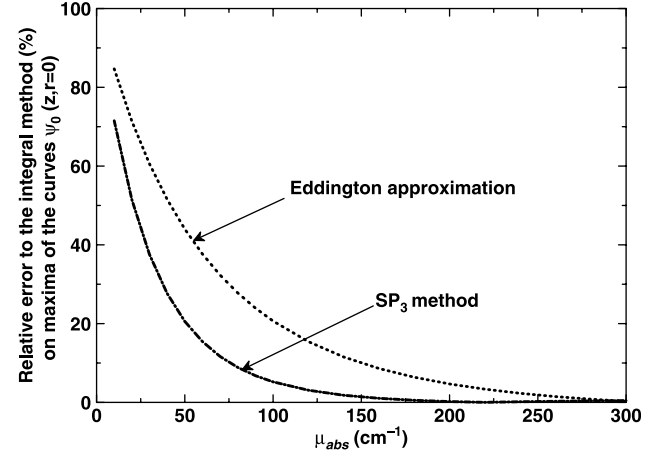


Figure 6. Relative error to the integral method on maxima of $\Psi_0(z, r = 0)$ as a function of the value of the monochromatic absorption coefficient. Dotted-dashed line: SP₃ method; dotted line: Eddington approximation.

decreases, figures 4 and 5 show that differences appear between the results obtained using different methods. More precisely, figures 4 and 5 show that for absorption coefficient values of 100 and 10 cm^{-1} , the results obtained using the direct method remain very close to those obtained with the integral method, while those obtained with Eddington and SP₃ methods underestimate the maximum value of the isotropic part of the distribution function and overestimate the values of the function on the wings of the profile. However, it is important to note that, as expected, the results obtained by the SP₃ model are closer to those obtained with the reference integral model than those obtained with the Eddington model. Figures 3–5 clearly show that for all values of the absorption coefficient, the results obtained using the FVM and the reference integral method are in very close agreement. Then, we have calculated the relative errors of the results obtained with the SP₃ and Eddington methods on the calculation of the maximum of $\Psi_0(z, r = 0)$, taking the integral method as a reference. Figure 6 shows the evolution of these relative errors as a function of the value of the monochromatic absorption coefficient. As expected, the relative error of the SP₃ method is less than the one of the Eddington approximation for all values of the absorption coefficient. Both errors decrease as μ_{abs} increases and then, as expected, for high values of the absorption coefficient we note that the Eddington and SP₃ methods become as accurate as the integral method.

Figure 7 shows the evolution of the computation times of the different methods as a function of the value of the monochromatic absorption coefficient. It is important to note that the approximate models (i.e. Eddington and SP₃ methods) require to solve elliptic equations (equation (18) and equations (19) and (20), respectively). In this work, to solve elliptic equations, we have used the iterative resolution method based on the D03EBF module of the library NAG Fortran. The resolution of the RTE based on the FVM and the classic integral methods do not require interactions. Consequently, figure 7 shows that the computation time of results obtained using the direct FVM is independent of the value of the

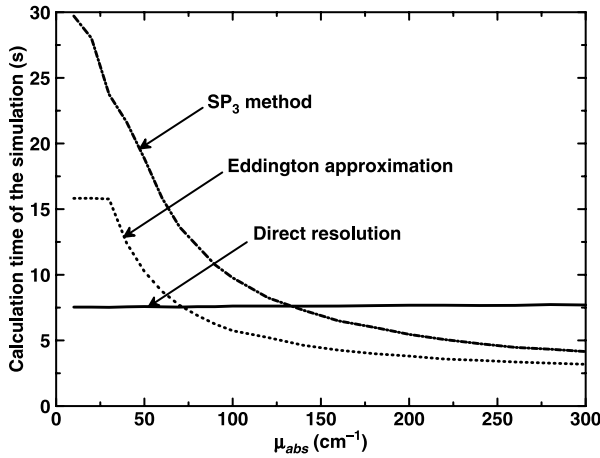


Figure 7. Computation times of the Eddington, SP₃ methods and the FVM as a function of the monochromatic absorption coefficient. Dotted–dashed line: SP₃ method; solid line: direct FVM; dotted line: Eddington approximation.

absorption coefficient. This is also the case of the integral method. However, without any optimization of the calculation procedure (as those mentioned in section 2.3), computation times of results obtained using the integral method are 10 000 times longer than those obtained with the FVM. Therefore, computation times of the classic integral model are not shown in figure 7. Conversely, we note that the computation times of the simulations carried out using SP₃ and Eddington methods depend on the value of the absorption coefficient and decrease when μ_{abs} increases. It is interesting to note that for the values of the absorption coefficient greater than 70 cm^{-1} for the Eddington approximation and 135 cm^{-1} for the SP₃ method, the approximate models become faster than the direct FVM. In conclusion of the results obtained for a monochromatic case, we have shown that whatever the value of the absorption coefficient, the results obtained using the direct FVM are in excellent agreement with the reference integral model for a significantly reduced computation time. When the absorption coefficient is high enough, Eddington and SP₃ methods are as accurate and become faster than the FVM. However, when the absorption coefficient decreases, the approximate methods become less accurate and more computationally expensive than the direct FVM.

3.2. Propagation of a double-headed streamer

In this section, we report and compare modelling results on a double-headed streamer developing at ground pressure (760 Torr) obtained with different photoionization models. It is important to mention that in this paper, we consider only monochromatic cases for photoionization. In the previous section, we have shown that the direct method is in very good agreement with the reference integral model for all values of the absorption coefficient and that, as expected, the SP₃ model is more accurate than the Eddington model. Then in this section, we propose to compare results obtained using the direct method and the SP₃ model, considering the direct method as the reference.

The geometry of the simulation domain is the same as in (Liu and Pasko 2004 (figure 3)). Two remote electrodes with a certain potential difference establish a high uniform Laplacian field $E_0 = 48 \text{ kV cm}^{-1}$ ($> E_k$, the breakdown threshold field). All results presented in this paper are obtained assuming a total neutral density of $N_0 = 2.688 \times 10^{19} \text{ cm}^{-3}$ and therefore $E_0/N_0 = 178.6 \text{ Td}$ ($1 \text{ Td} = 10^{-17} \text{ V cm}^2$). Under the influence of this applied field, a double-headed streamer is launched by placing a neutral plasma cloud in the simulation domain. The initial plasma cloud has a Gaussian distribution in space:

$$n_e(r, z)|_{t=0} = n_p(r, z)|_{t=0} = n_0 \exp \left[-\left(\frac{r}{\sigma_r} \right)^2 - \left(\frac{z - z_0}{\sigma_z} \right)^2 \right]. \quad (43)$$

The centre of the Gaussian distribution is located in the middle of the simulation domain, at $z_0 = 0.7 \text{ cm}$, and it is assumed that $\sigma_r = \sigma_z = 0.02 \text{ cm}$ and $n_0 = 10^{14} \text{ cm}^{-3}$. The size of the computational domain is $1.4 \text{ cm} \times 0.05 \text{ cm}$. The computational grid is uniform in both radial and axial directions. As part of the preparatory work for the model studies presented in this paper, we have conducted several test runs with different numbers of grid points with uniform meshes. In this paper, for the direct method we have used a total number of cells of $N_z \times N_r = 4000 \times 200$ and $N_\theta = N_\phi = 14$. For the SP₃ model, we have used a total number of cells of $N_z \times N_r = 3000 \times 200$. We have checked that the results obtained with the different methods with these meshes are the same as those obtained with more refined grids. Finally, it is important to note that this test case is very close to the one studied in Liu and Pasko (2004) and Bourdon *et al* (2007). The main difference lies in the fact that in this paper, we consider only monochromatic absorption coefficients and we do not take into account the spectral dependence of photoionization as in Liu and Pasko (2004) and Bourdon *et al* (2007). Small other differences can also be noted on the simulation domain and on the grid. Indeed, in this paper, the radial expansion of the domain is reduced to 0.05 cm (instead of 0.125 cm) and the number of grid points has been slightly increased.

In this work, the charged species transport equations are solved using a flux-corrected transport (FCT) method (Bourdon *et al* (2007) and references therein). The third order QUICKEST scheme is used as the high order scheme and an upwind scheme for the low order scheme. The flux limiter derived by Zalesak (1979) is adopted for this FCT method. The finite difference form of Poisson's equation is solved using the D03EBF module of the NAG Fortran library (<http://www.nag.co.uk>). For the test studies presented in this paper, all transport parameters and reaction rates in air are taken from (Morrow and Lowke 1997). The boundary conditions and the definition of the time steps are the same as in Bourdon *et al* (2007).

Finally, to compute the photoionization source term with a monochromatic approach, we consider in this paper the same parameters as in Ségur *et al* (2006) in equations (12), (13) and (14). That is to say, we assume that $p_q = 30 \text{ Torr}$, $p = 760 \text{ Torr}$ and $\xi^{\text{ph}} = 0.0554$ and

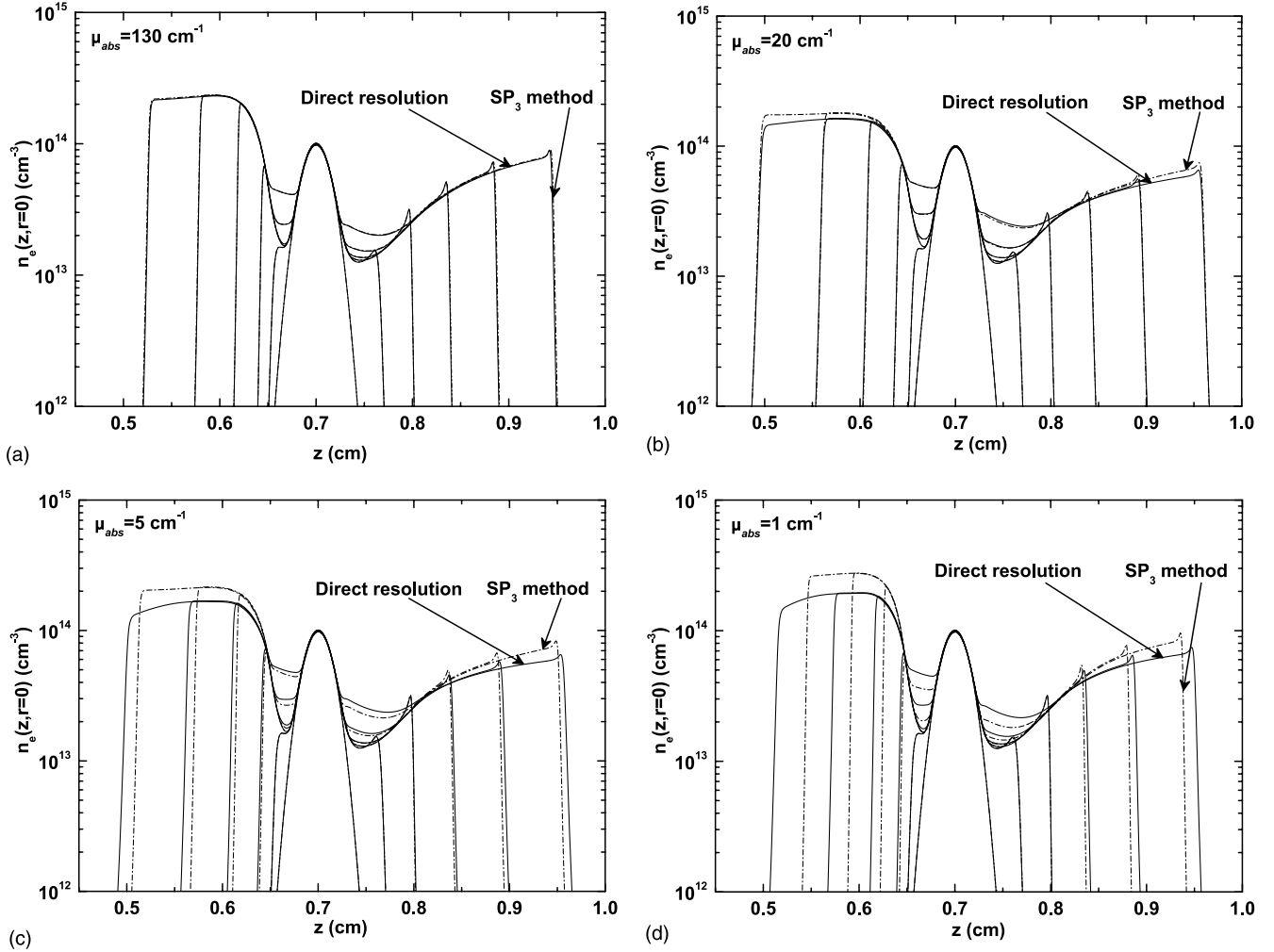


Figure 8. Electron density profiles on the symmetry axis ($r = 0$) of the computational domain at various time moments calculated using the SP₃ model and the direct FVM to solve the RTE with monochromatic absorption coefficients $\mu_{\text{abs}} = 130, 20, 5$ and 1 cm^{-1} . Dotted–dashed line: SP₃ method; solid line: direct FVM. Results are shown for different time moments ranging from 0 to 2.5 ns, with a timestep of 0.5 ns.

$v_u/v_i = 0.6$. It is important to note that these parameters have been determined in [Séguir *et al* \(2006\)](#) to obtain with a monochromatic approach (i.e. for $\mu_{\text{abs}} = 130 \text{ cm}^{-1}$) a good agreement at low values of R with the reference model of [Zheleznyak *et al* \(1982\)](#) for photoionization in air. In [Bourdon *et al* \(2007\)](#), we have shown that to obtain a very good agreement with the reference model of [Zheleznyak *et al* \(1982\)](#) for photoionization in air in a wide range of distances $0.1 < p_{\text{O}_2} R < 150 \text{ Torr cm}$ (where p_{O_2} is the partial pressure of molecular oxygen), it is necessary to use a three-exponential fit. The three corresponding monochromatic absorption coefficients are 6, 16 and 89 cm^{-1} . In this paper, we use the values of the parameters (p_q , p and ξ^{ph} and v_u/v_i) determined in [Séguir *et al* \(2006\)](#) and we keep them constant whatever the value of μ_{abs} . In this section, to test the photoionization models, values of the absorption coefficient μ_{abs} ranging from 1 to 130 cm^{-1} are used. Figure 8 compares the electron number density distributions on the symmetry axis of the computational domain calculated using the SP₃ and the direct finite volume models for four values of the absorption coefficient $\mu_{\text{abs}} = 130, 20, 5$ and 1 cm^{-1} . The results are shown for the moments of time from $t = 0$ to

$t = 2.5 \text{ ns}$, with a timestep of 0.5 ns. For all values of the absorption coefficient, figure 8 shows that a positive streamer propagates from the left-hand side wing of the initial plasma cloud towards decreasing values of z and a negative streamer propagates from the right-hand side wing of the initial plasma cloud towards increasing values of z . For $\mu_{\text{abs}} = 130 \text{ cm}^{-1}$, we note that there is an excellent agreement between the results obtained with the direct and SP₃ models for both streamers. As the value of μ_{abs} decreases, figure 8 shows that discrepancies between the results obtained with the direct and SP₃ models increase, and these differences increase as the streamers propagate. More precisely, a fairly good agreement between both models is obtained for $\mu_{\text{abs}} = 20 \text{ cm}^{-1}$, but significant differences are obtained for $\mu_{\text{abs}} = 5$ and 1 cm^{-1} . As the value of μ_{abs} decreases, the SP₃ model overestimates the electron density in both streamer heads and streamers propagate more slowly than those calculated using the direct method. It is interesting to point out that this effect is more pronounced on the positive streamer side. For the very small value of $\mu_{\text{abs}} = 1 \text{ cm}^{-1}$, we note that the results obtained using the SP₃ model overestimate by a factor of 2 the electron density in both streamer heads at $t = 2.5 \text{ ns}$.

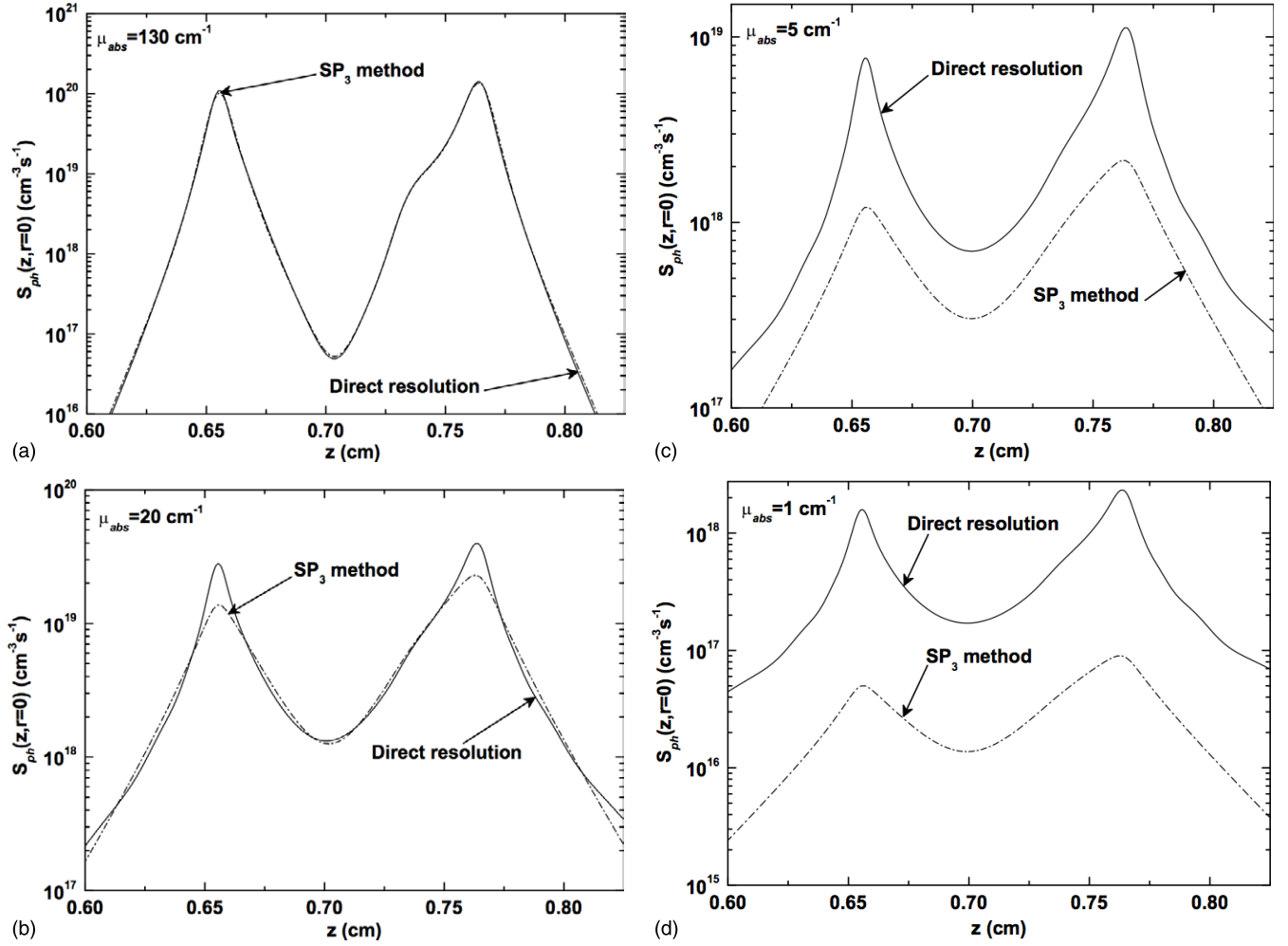


Figure 9. Axial profiles of the photoionization source term $S_{ph}(z, r = 0)$ on the symmetry axis at $t = 0.5$ ns with $\mu_{abs} = 130, 20, 5$ and 1 cm^{-1} . Dotted–dashed line: SP_3 method; solid line: direct FVM.

Figure 9 compares the photoionization source terms on the symmetry axis of the computational domain calculated using the SP_3 and the direct method at $t = 0.5$ ns and for four values of the absorption coefficient $\mu_{abs} = 130, 20, 5$ and 1 cm^{-1} . As expected from figure 8, for $\mu_{abs} = 130 \text{ cm}^{-1}$, an excellent agreement is obtained between the photoionization source term calculated with the SP_3 model and the FVM. As the value of μ_{abs} decreases, figure 9 shows that discrepancies between the results obtained with the direct and SP_3 models increase. More precisely, we note that as the value of μ_{abs} decreases, the photoionization source term calculated using the SP_3 model is underestimated in both streamers in comparison with the results obtained using the direct method. For the very small value of $\mu_{abs} = 1 \text{ cm}^{-1}$, the SP_3 model underestimates photoionization source term in both streamers by nearly one order of magnitude at $t = 0.5$ ns. As mentioned in Bourdon *et al* (2007), the double-headed streamer is a high field test case and then, very rapidly as the streamer starts to propagate, the ionization term becomes stronger than the photoionization term everywhere. Then a one order difference observed for $\mu_{abs} = 1 \text{ cm}^{-1}$ on the photoionization source term has only a limited influence on the propagation of both streamers.

Finally, we propose to use the results presented in figures 8 and 9 to discuss the influence of the value of the absorption coefficient μ_{abs} on positive and negative streamer propagations. In the monochromatic photoionization models used in this paper, it is important to recall that the absorption coefficient μ_{abs} is in the RTE (equation (11)) and then has a direct influence on the value of $\Psi_0(\vec{r})$, but μ_{abs} is also in equation (14) giving $S_{ph}(\vec{r})$. As already mentioned, we note clearly in figure 8 that the influence of photoionization on positive streamer propagation is more significant than on the negative streamer one. As the value of μ_{abs} decreases from 130 to 20 cm^{-1} , the velocity of the negative streamer, calculated using the FVM for photoionization, increases and then remains nearly constant as μ_{abs} decreases down to 1 cm^{-1} . For the positive streamer, as the value of μ_{abs} decreases from 130 to 20 cm^{-1} , the velocity of the positive streamer calculated using the FVM for photoionization increases and then remains nearly constant as μ_{abs} decreases down to 5 cm^{-1} . As the value of μ_{abs} decreases from 5 to 1 cm^{-1} , the velocity of the positive streamer decreases and finally we note that the positive streamer obtained for $\mu_{abs} = 1 \text{ cm}^{-1}$ is slightly faster than the positive streamer obtained for $\mu_{abs} = 130 \text{ cm}^{-1}$. Then, it is interesting to point out that when the absorption coefficient is

high ($\mu_{\text{abs}} > 130 \text{ cm}^{-1}$), the photons emitted by the streamer head are strongly absorbed close to their emission point. Then seed electrons are produced very close to the head which may finally prevent the propagation of the streamer, as a small distance is necessary to produce the avalanche towards the streamer head. Conversely, for low values of μ_{abs} , the photons emitted by the streamer head are only weakly absorbed by the gas and may propagate quite far from the head. Furthermore as μ_{abs} is also in equation (14), we note that, for low values of μ_{abs} , the photons are not very efficient to produce seed electrons. All these results clearly show that there is a range of optimal values of the monochromatic absorption coefficient ($5 \text{ cm}^{-1} < \mu_{\text{abs}} < 130 \text{ cm}^{-1}$) which allow a rapid propagation of the positive streamer.

4. Conclusions

In this paper, we develop a direct accurate numerical solution of the RTE based on a FVM and we apply it to the simulation of positive and negative streamer propagations. The results are compared with approximate solutions of the RTE (Eddington and SP₃ models) and the reference integral model. In this work, we have considered only monochromatic cases. We have conducted two test studies of the performance of the different models to calculate the photoionization source term: a Gaussian emission source and a double-headed streamer developing in a strong uniform electric field (greater than the conventional breakdown field).

Our studies using the Gaussian radiation source have demonstrated that for all values of the absorption coefficient, the results obtained using the direct FVM are in excellent agreement with the reference integral model for a significantly reduced computation time. When the absorption coefficient is high enough, Eddington and SP₃ methods are as accurate and become faster than the direct method. However, when the absorption coefficient decreases, approximate methods become less accurate and more computationally expensive than the direct FVM.

Our studies of the double-headed streamer have demonstrated that when the absorption coefficient is high enough ($\mu_{\text{abs}} = 130 \text{ cm}^{-1}$), the results obtained using SP₃ and direct methods are in excellent agreement. As the value of μ_{abs} decreases, discrepancies between the results obtained with the direct and SP₃ models increase, and these differences increase as the streamers advance. For low values of μ_{abs} , the SP₃ model overestimates the electron density in both streamers and positive and negative streamers propagate more slowly than those calculated using the direct method. Furthermore, as the value of μ_{abs} decreases, the SP₃ model underestimates the photoionization source term in both streamers in comparison with the direct method. For the very low value of $\mu_{\text{abs}} = 1 \text{ cm}^{-1}$, the SP₃ model underestimates the photoionization source term in both streamers by nearly one order of magnitude at $t = 0.5 \text{ ns}$. The corresponding electron density calculated with the SP₃ model overestimates by a factor of 2 the electron density in both streamer heads calculated with the direct FVM at $t = 2.5 \text{ ns}$. For the double-headed streamer with a strong uniform applied field, photoionization only plays

a role during the very early stage of the development of the streamer. Then, a one order difference observed for $\mu_{\text{abs}} = 1 \text{ cm}^{-1}$ on the photoionization source term has only a limited influence on the propagation of both streamers. For positive streamers propagating in a weak external field (less than the conventional breakdown field) as in point-to-plane geometry, the photoionization source term dominates the ionization source term in most of the region ahead of the streamer during its propagation and then, for low values of the absorption coefficient, significant differences between the streamers calculated using the FVM and the SP₃ model are expected. The related results will be presented in a separate dedicated paper.

Finally, in this work, we have studied the direct FVM for monochromatic cases. To accurately simulate streamers propagating in air at atmospheric pressure, it is necessary to take into account the spectral dependence of the photoionization process. In Bourdon *et al* (2007), we have developed an accurate and efficient method to do that based on a three-group approach. This approach could be very easily used with the FVM developed in this paper to simulate streamer propagation in air in high and weak fields. The related results will be presented in a separate dedicated paper.

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